第 4 阶段。重新定位和停用节点

AFF and FAS Controller Upgrade

NetApp
February 03, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap-systems-upgrade/upgrade-arl-auto/stage_4_index.html on February 03, 2022. Always check docs.netapp.com for the latest.

目录

第 4 阶段。重新定位和停用节点 1
第 4 阶段。重新定位和停用节点 2 1
将非根聚合和 NAS 数据 LIF 从 node2 重新定位到 node3 1
停用 node2 3

第 4 阶段。重新定位和停用节点

第 4 阶段。重新定位和停用节点 2

在阶段 4 中，所有非根聚合将从 node2 重新定位到 node3； node2 拥有的非 SAN 数据 LIF 将移至 node3 。您必须记录必要的 node2 信息，然后停用 node2 。

步骤

1. "将非根聚合和 NAS 数据 LIF 从 node2 重新定位到 node3"
2. "停用 node2"

将非根聚合和 NAS 数据 LIF 从 node2 重新定位到 node3

在将 node2 替换为 node4 之前，必须将 node2 所拥有的非根聚合重新定位到 node3 。

上一阶段的后处理检查完成后， node2 的资源释放将自动启动。非根聚合和非 SAN 数据 LIF 将从 node2 迁移到 node3 。

在升级操作步骤期间，远程 LIF 处理 SAN LUN 的流量。升级期间，集群或服务运行状况无需移动 SAN LIF 。

迁移聚合和 LIF 后，此操作将暂停以进行验证。在此阶段，您必须验证是否所有非根聚合和非 SAN 数据 LIF 都已迁移到 node3 。

> 不会修改聚合和 LIF 的主所有者；只会修改当前所有者。

步骤

1. 验证所有非根聚合是否均已联机及其在 node3 上的状态：

   ```
   storage aggregate show -node node3 -state online -root false
   ```

   以下示例显示 node2 上的非根聚合处于联机状态：

   ```
   cluster::> storage aggregate show -node node3 state online -root false

   Aggregate      Size         Available   Used%   State   #Vols  Nodes
   RAID     Status
   ----------     ---------    ---------   ------  -----   -----  ------
   -------  ------
   aggr_1         744.9GB      744.8GB      0%     online  5      node2
   raid_dp  normal
   aggr_2         825.0GB      825.0GB      0%     online  1      node2
   raid_dp  normal
   2 entries were displayed.
   ```

   如果 node3 上的聚合脱机或变为外部聚合，请在 node3 上使用以下命令将其联机，每个聚合一次：

```
storage aggregate online -aggregate <aggr_name>
```

2. 在 node3 上使用以下命令并检查输出，以验证 node3 上的所有卷是否联机：

   ```
   volume show -node <node3> -state offline
   ```

   如果 node3 上的任何卷脱机，请在 node3 上使用以下命令将其联机，每个卷一次：

   ```
   volume online -vserver <vserver_name> -volume <volume_name>
   ```

   要与此命令结合使用的 ` <vserver_name>` 可在上一个 `volume show` 命令的输出中找到。

3. 验证 LIF 是否已移至正确的端口且状态为 `up` 。如果任何 LIF 已关闭，请为每个 LIF 输入以下命令，将 LIF 的管理状态设置为 `up` ：

   ```
   network interface modify -vserver <vserver_name> - lif <LIF_name> -home-node
   <node_name> - status-admin up
   ```

4. 如果新硬件上不存在当前托管数据 LIF 的端口，请将其从广播域中删除：

   ```
   network port broadcast-domain remove-ports
   ```

5. 【第 5 步】输入以下命令并检查输出，验证 node2 上是否没有剩余数据 LIF ：

   ```
   network interface show - curr-node <node2>-role data
   ```

6. 如果配置了接口组或 VLAN ，请完成以下子步骤：

   a. 记录 VLAN 和接口组信息，以便在启动 node3 后在 node3 上重新创建 VLAN 和接口组。

   b. 从接口组中删除 VLAN ：

      ```
      network port vlan delete -node <nodename > -port <ifgrp> -vlan-id <vlan_ID>
      ```

   c. 输入以下命令并检查其输出，以检查节点上是否配置了任何接口组：

      ```
      network port ifgrp show -node <node2>-ifgrp <ifgrp_name> -instance
      ```

      系统将显示节点的接口组信息，如以下示例所示：

```
cluster::> network port ifgrp show -node node2 -ifgrp a0a -instance
                 Node: node3
 Interface Group Name: a0a
Distribution Function: ip
        Create Policy: multimode_lacp
          MAC Address: 02:a0:98:17:dc:d4
   Port Participation: partial
        Network Ports: e2c, e2d
             Up Ports: e2c
           Down Ports: e2d
```

a. 如果节点上配置了任何接口组，请记录这些组的名称以及分配给它们的端口，然后输入以下命令删除这些端口，每个端口输入一次：

```
network port ifgrp remove-port -node <nodename > -ifgrp <ifgrp_name> -port
<netport>
```

停用 node2

要停用 node2 ，您需要正确关闭 node2 并将其从机架或机箱中卸下。

步骤

1. 恢复操作：

   s系统控制器更换恢复

   节点会自动暂停。

升级完成后，您可以停用 node2 。请参见 "停用旧系统"。

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.